Q1: 16 March (Shift 1) - Single Correct

Which of the following Boolean expression is a tautology ?

(1) $(p \wedge q) \vee (p \vee q)$

(2) $(p \wedge q) \vee (p \rightarrow q)$

(3) $(p \wedge q) \wedge (p \rightarrow q)$

(4) $(p \wedge q) \rightarrow (p \rightarrow q)$

Q2: 17 March (Shift 1) - Single Correct

If the Boolean expression $(p \Rightarrow q) \Leftrightarrow (q * (\sim p))$ is a tautology, then the Boolean expression $p * (\sim q)$ is equivalent to

(1) $q \Rightarrow p$

(2) $\sim q \Rightarrow p$

(3) $p \Rightarrow \sim q$

(4) $p \Rightarrow q$

Q3: 17 March (Shift 2) - Single Correct

If the Boolean expression $(p \wedge q) \circledast (p \otimes q)$ is a tautology, then $\circledast$ and $\otimes$ are respectively given by

(1) $\rightarrow, \rightarrow$

(2) $\wedge, \vee$

(3) $\vee, \rightarrow$

(4) $\wedge, \rightarrow$

Q4: 18 March (Shift 2) - Single Correct

If P and Q are two statements, then which of the following compound statement is a tautology?

(1) $((P \Rightarrow Q) \wedge \sim Q) \Rightarrow Q$

(2) $((P \Rightarrow Q) \wedge \sim Q) \Rightarrow \sim P$

(3) $((P \Rightarrow Q) \wedge \sim Q) \Rightarrow P$

(4) $((P \Rightarrow Q) \wedge \sim Q) \Rightarrow (P \wedge Q)$

Answer Key

Q1 (4) **Q2** (1) **Q3** (1) **Q4** (2)

Q1 (4)

p	q	$p \wedge q$	$p \rightarrow q$	$(p \wedge q) \rightarrow (p \rightarrow q)$
T	T	T	T	T
T	F	F	F	T
F	T	F	T	T
F	F	F	T	T

$(p \wedge q) \rightarrow (p \rightarrow q)$ is tautology

Q2 (1)

$\because p \rightarrow q \equiv \sim p \vee q$

So, $* \equiv v$

Thus, $p^*(\sim q) \equiv pv(\sim q)$

$\equiv q \rightarrow p$

Q3 (1)

Option (1)

$(p \wedge q) \longrightarrow (p \rightarrow q)$

$= \sim (p \wedge q) \vee (\sim p \vee q)$

$= (\sim p \vee \sim q) \vee (\sim p \vee q)$

$= \sim p \vee (\sim q \vee q)$

$= \sim p \vee t$

$= t$

Option (2)

$(p \wedge q) \wedge (p \vee q) = (p \wedge q)(\text{ Not a tautology })$

Option (3) $(p \wedge q) \vee (p \rightarrow q)$

$= (p \wedge q) \vee (\sim p \vee q)$

$= \sim p \vee q$

(Not a tautology)

Option (4) $(p \wedge q) \wedge (p \rightarrow q)$

$= (p \wedge q) \wedge (\sim p \vee q)$

$= p \wedge q$

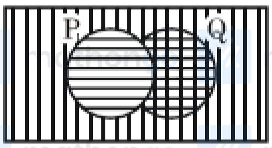
(Not a tautology)

Q4 (2)

$$\begin{aligned} &((P \rightarrow Q) \wedge \sim Q) \\ &\equiv (\sim P \vee Q) \wedge \sim Q \\ &\equiv (\sim P \wedge \sim Q) \vee (Q \wedge \sim Q) \\ &\equiv \sim P \wedge \sim Q \end{aligned}$$

LHS of all the options are some i.e.

$$\begin{aligned} &(A)\ (P \wedge \sim Q) \rightarrow Q \\ &\equiv \sim (\sim P \wedge \sim Q) \vee Q \\ &\equiv (P \vee Q) \vee Q \neq \text{ tautology} \\ &(B)\ (\sim P \wedge \sim Q) \rightarrow \sim P \\ &\equiv \sim (\sim P \wedge \sim Q) \vee \sim P \\ &\equiv (P \vee Q) \vee \sim P \end{aligned}$$



$\Rightarrow$ Tautology

(C) $(\sim P \wedge \sim Q) \rightarrow P$

$\equiv (P \vee Q) \vee P \neq$ Tautology

(D) $(\sim P \wedge \sim Q) \rightarrow (P \wedge Q)$

$\equiv (P \vee Q) \vee (P \wedge Q) \neq$ Tautology

Aliter:

P	Q	$P \vee Q$	$P \vee Q$	$\sim P$	$(P \vee Q) \vee \sim P$
T	T	T	T	F	T
T	F	T	F	F	T
F	T	T	F	T	T
F	F	F	F	T	T